



Observability Day

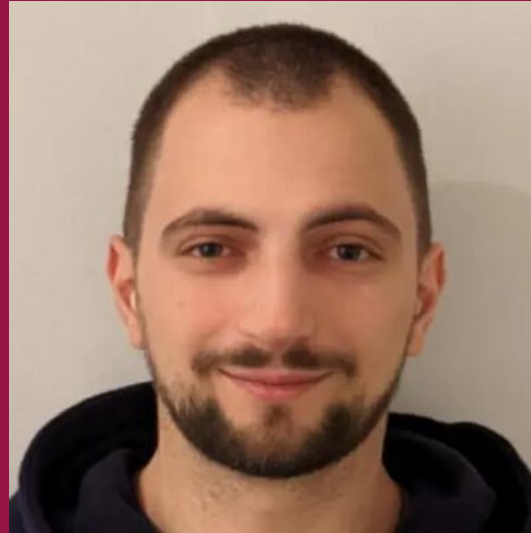
EUROPE

A Practical Blueprint for Evolving Observability Schemas

*Benedikt Bongartz, Red Hat
& Sebastian Rabenhorst, Shopify*

Who We Are

Observability Day
EUROPE



Benedikt Bongartz
Principal Software Engineer
Red Hat

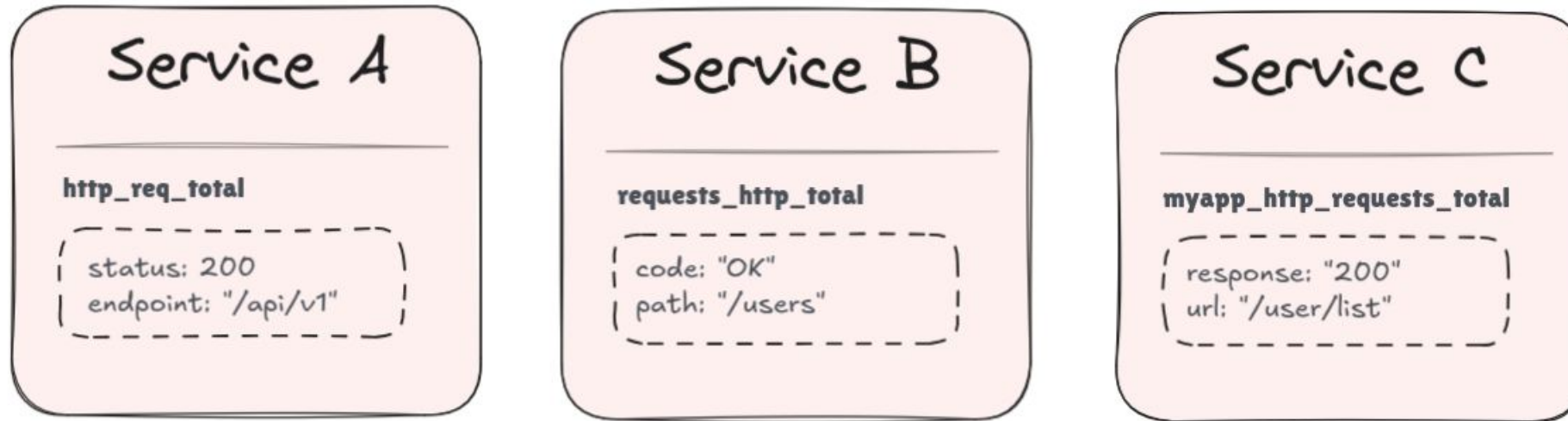


Sebastian Rabenhorst
Staff Engineer Observability
Shopify

- Schemas & Semantic Conventions
- OpenTelemetry SemConv evolution
- Changes and challenges
- Techniques and trade-offs
- Shopify's approach
- Q&A

Schemas and Semantic Conventions

Why Do We Need Schemas?



```
>_ sum(rate(http_req_total{status="200"}[5m])) +  
    sum(rate(requests_http_total{code="OK"}[5m])) +  
    sum(rate(myapp_http_requests_total{response="200"}[5m]))  
                                         RPS
```

HTTP server

Metric: `http.server.request.duration`

This metric is required.

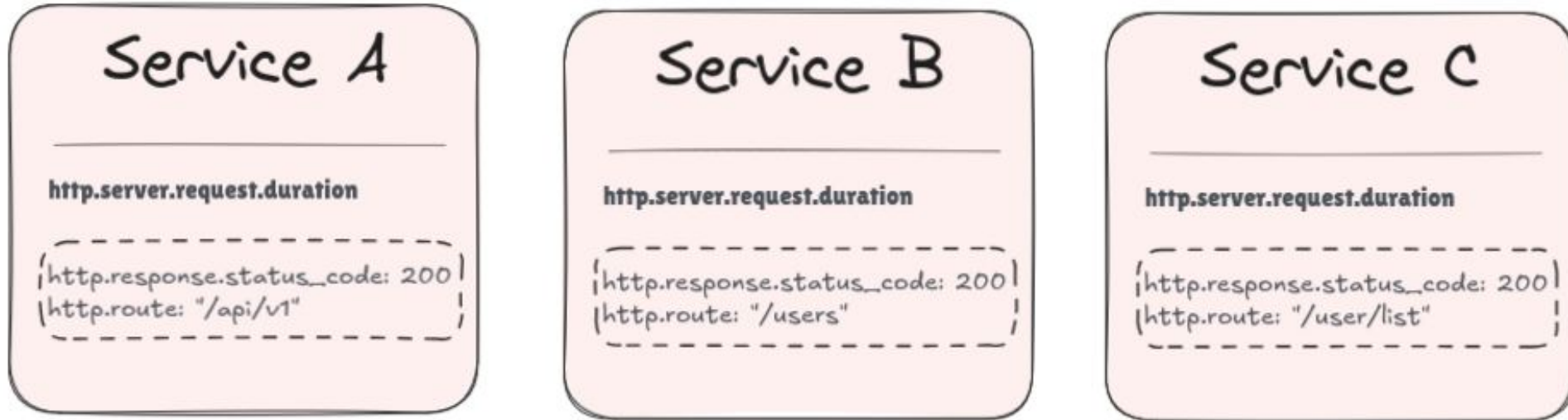
When this metric is reported alongside an HTTP server span, the metric value SHOULD be the same as the HTTP server span duration.

This metric SHOULD be specified with [ExplicitBucketBoundaries](#) of [0.005, 0.01, 0.025, 0.05, 0.075, 0.1, 0.25, 0.5, 0.75, 1, 2.5, 5, 7.5, 10].

Name	Instrument Type	Unit (UCUM)	Description	Stability	Entity Associations
<code>http.server.request.duration</code>	Histogram	s	Duration of HTTP server requests.	stable	

Attributes:

Key	Stability	Requirement Level	Value Type	Description	Example Values
http.request.method	stable	Required	string	HTTP request method. [1]	GET ; POST ; HEAD
url.scheme	stable	Required	string	The URI scheme component identifying the used protocol. [2]	http ; https
error.type	stable	Conditionally Required If request has ended with an error.	string	Describes a class of error the operation ended with. [3]	timeout ; java.net.UnknownHostException ; server_certificate_invalid ; 500
http.response.status_code	stable	Conditionally Required If and only if one was received/sent.	int	HTTP response status code .	200
http.route	stable	Conditionally Required If and only if it's available	string	The matched route template for the request. This MUST be low-cardinality and include all static path segments, with dynamic path segments represented with placeholders. [4]	/users/:userID? ; my-controller/my-action/{id?}



```
>_ sum(rate(http_server_request_duration_seconds_count{  
    http_response_status_code="200"}[5m]))
```

RPS

SemConv Evolution

Breaking Changes per Version (1.4.0 - 1.39.0)



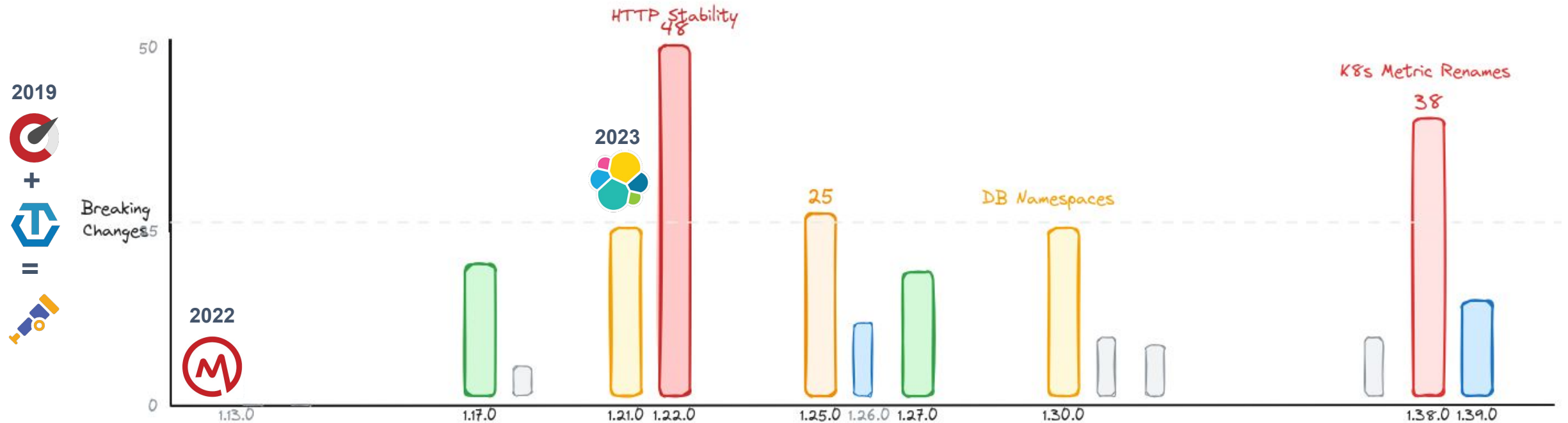
Key Events

- 1.22.0 (48 changes): HTTP conventions stabilized
- 1.38.0 (38 changes): K8s metrics namespace reorganization
- 1.30.0 (23 changes): Database attributes moved to vendor

Total: 37 versions tracked
Breaking Changes: 270+ renames
(9 versions, ~160 changes)

SemConv Evolution

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Do Schemas Make Life Simpler?

HTTP server duration metric

Metric changes:

- **Name:** `http.server.duration` → `http.server.request.duration`
- **Unit:** `ms` → `s`
- **Description:** Measures the duration of inbound HTTP requests. → Duration of HTTP server requests.

• **Histogram buckets:** boundaries updated to reflect change from milliseconds to seconds, and zero bucket boundary removed

- **Attributes:** see table below

Attribute change	Comments
<code>http.route</code>	No change
<code>http.method</code> → <code>http.request.method</code>	Now captures only 9 common HTTP methods by default plus <code>_OTHER</code>
<code>http.status_code</code> → <code>http.response.status_code</code>	
<code>http.scheme</code> → <code>url.scheme</code>	Now factors in X-Forwarded-Proto span , Forwarded#proto span headers
<code>net.protocol.name</code> → <code>network.protocol.name</code>	Recommended → Conditionally required if not <code>http</code> and <code>network.protocol.version</code> is set
<code>net.protocol.version</code> → <code>network.protocol.version</code>	Examples fixed: <code>2.0</code> → <code>2</code> and <code>3.0</code> → <code>3</code> ; see note if migrating from <= v1.19.0
<code>net.host.name</code> → <code>server.address</code>	• Recommended → Opt-In (due to high-cardinality vulnerability since based on HTTP headers) • Now based only on Host span, :authority span, X-Forwarded-Host span, Forwarded#host span headers
<code>net.host.port</code> → <code>server.port</code>	• Recommended → Opt-In (due to high-cardinality vulnerability since based on HTTP headers) • Now based only on Host span, :authority span, X-Forwarded-Host span, Forwarded#host span headers
New: <code>error.type</code>	

References:

- [Metric `http.server.duration` v1.20.0](#)
- [Metric `http.server.request.duration` v1.23.1 \(stable\)](#)

```
file_format: 1.1.0
schema_url: https://opentelemetry.io/schemas/1.39.0
versions:
  1.39.0:
    all:
      changes:
        - rename_attributes:
            attribute_map:
              linux.memory.slab.state: system.memory.linux.slab.state
              peer.service: service.peer.name
```

```
...
1.22.0:
  metrics:
    changes:
      # https://github.com/open-telemetry/semantic-conventions/pull/224
      - rename_metrics:
          http.client.duration: http.client.request.duration
          http.server.duration: http.server.request.duration
...
1.21.0:
  spans:
    changes:
      # https://github.com/open-telemetry/opentelemetry-specification/pull/3355
      - rename_attributes:
          attribute_map:
            http.method: http.request.method
            http.status_code: http.response.status_code
            http.scheme: url.scheme
            http.url: url.full
            http.request_content_length: http.request.body.size
            http.response_content_length: http.response.body.size
```

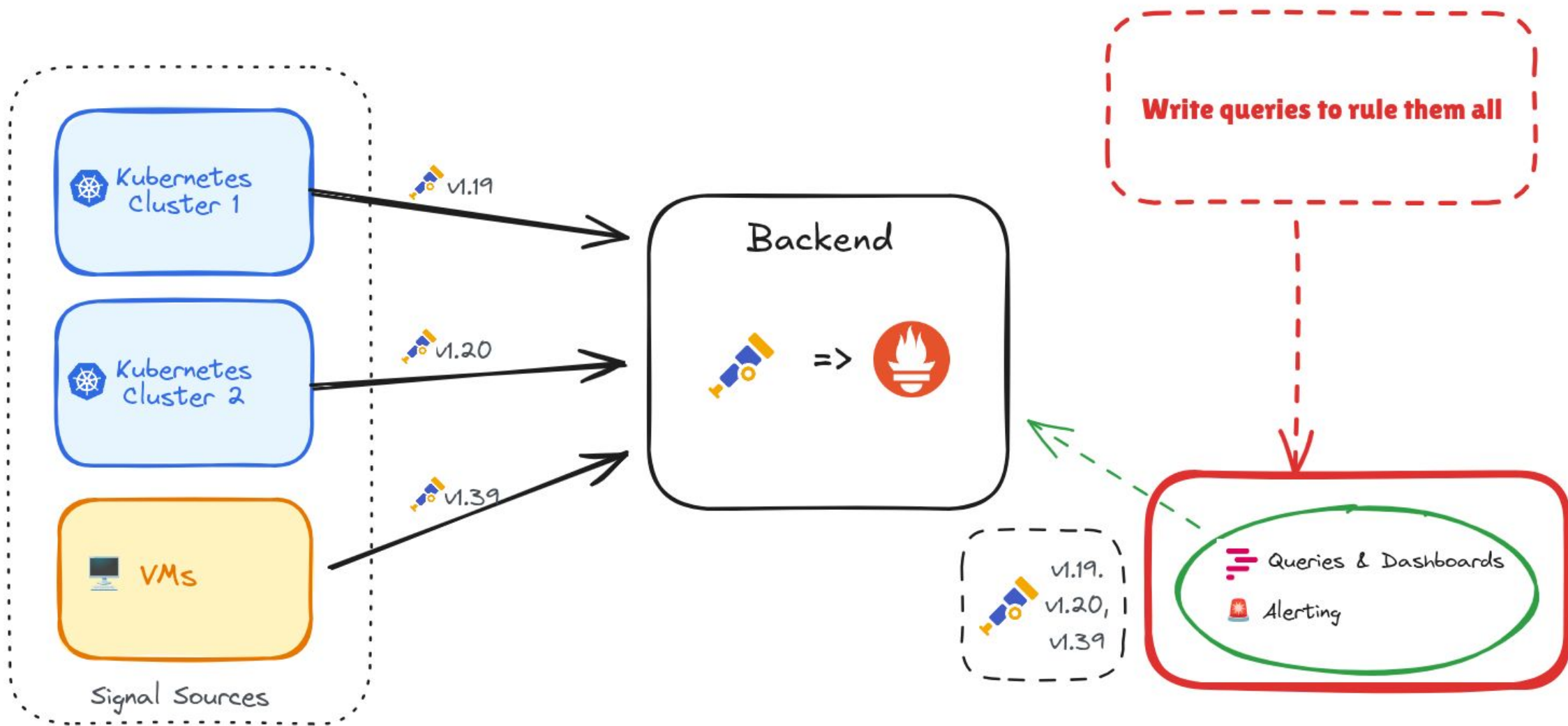
Do Schemas Improve Collaboration?

TL;DR: They Do!

- Consistent telemetry
- Predictable cardinality
- Reliable dashboards and queries
- Safer instrumentation changes
- Automated validation and enforcement across the team/company



Breaking Changes - Query



Breaking Changes - Query

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promql: Implement support for versioned OTel semantic conventions #17702

Draft aknuds1 wants to merge 5 commits into `prometheus:main` from `aknuds1:arve/otlp-semconv-schema`

Conversation 0 Commits 5 Checks 31 Files changed 21 +2,887 -0

aknuds1 commented on Dec 16, 2025 • edited

Which issue(s) does the PR fix:

Implement support for versioned OTel semantic conventions, with automatic querying for all metric versions. Behind feature flag "semconv-versioned-read" - this is a working title, final name is TODO :)

TODO:

- ☐ Transform the label names of all series belonging to other versions to correspond to the queried version
- ☐ Compute semantic conventions group IDs (how?) - potentially unnecessary with next version of semantic conventions format
- ☐ Use OTel schema format v2, where definitions are included (@jsuereth knows)
- ☐ Decide on feature flag name
- ☐ Resolve TODOs in code

Does this PR introduce a user-facing change?

[FEATURE] Add support for versioned OTel semantic conventions, with automatic querying for all metric ver

aknuds1 force-pushed the `arve/otlp-semconv-schema` branch 2 times, most recently from `26095a8` to `c6f01ec` last month

Reviewers: bwplotka

Assignees: No one assigned

Labels: `area/opentelemetry`, `component/promql`, `kind/feature`

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues.

How To Rename Metrics Without Impacting Somebody's Observability

Bartłomiej Płotka, Google & Arianna Vespri, Independent

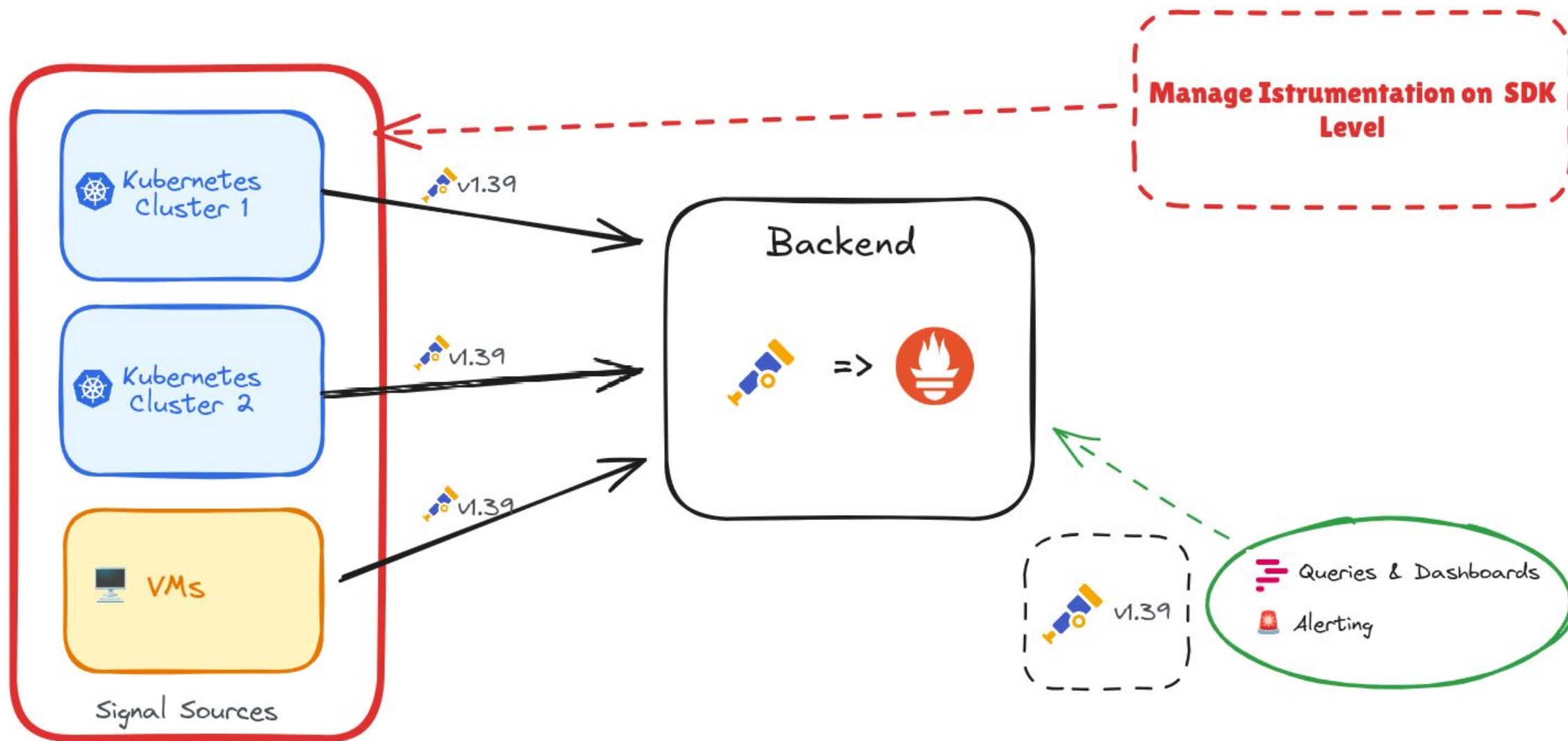
<https://www.youtube.com/watch?v=Rw4c7lmdyFs>

<https://github.com/prometheus/prometheus/pull/17702>

Average request duration

```
>_ (sum(rate(http_server_request_duration_seconds_sum[5m])))  
or  
    (sum(rate(http_server_duration_seconds_sum[5m])) / 1000)
```

Breaking Changes - Instrumentation



SemConv Migrations

Instrumentation capabilities #2542

Open



TylerHelmuth opened on Jan 18, 2024 · edited by TylerHelmuth

Edits Member ...

Component(s)

instrumentation

Describe the issue you're reporting

The [HTTP semantic conventions are now stable](#) and they bring lots of impactful breaking changes. As OTel languages take the stable conventions by default our auto-instrumentation images will get updated and those breaking changes will make their way to our users.

Unlike users who manually add instrumentation libraries, the Operator currently adds a layer of obfuscation around what auto-instrumentation is actually installed. In situations where the operator is installing an "agent", like for Java, .Net, or Go, there is an extra layer of obfuscation.

We need to be very careful with these updates so that we do not causes our users to upgrade unknowingly, causing their telemetry to change. Here are some of the solutions we discussed on the SIG meeting on 2023-01-18:

- Determine a migration plan, per language, for auto-instrumentation users. Things we can do to help:
 - Documentation
 - Warnings in the webhook
 - Feature gates
 - Major version bumps of our images.
- Determine how long we will wait, per language, before setting the default language image to use instrumentation libraries with the breaking changes.
- Communicate concerns with the TC/GC via the Maintainers meeting

Create sub-issue

TylerHelmuth added `needs triage` `area:auto-instrum...` `auto-instrumentat...` `auto-instrumentat...` `auto-instrumentat...` `auto-instrumentat...` `auto-instrumentat...` `auto-instrumentat...` and removed `needs triage` on Jan 18, 2024

TylerHelmuth pinned this issue on Jan 18, 2024

<https://github.com/open-telemetry/opentelemetry-operator/issues/2542>
<https://opentelemetry.io/docs/specs/semconv/non-normative/http-migration/>

Assignees

No one - [Assign yourself](#)

Labels

`area:auto-instrumentation`
`auto-instrumentation:apache-httpd`
`auto-instrumentation:dotnet`
`auto-instrumentation:go`
`auto-instrumentation:java`
`auto-instrumentation:nginx`
`auto-instrumentation:nodejs`
`auto-instrumentation:python`

Type

No type

Projects

Release instructions

Steps to release a new version of the OpenTelemetry Operator:

1. Create a `Prepare release x.y.z` pull request with the following content:
 - i. Set the version you're releasing as an environment variable for convenience: `export VERSION=0.n+1.0`
 - ii. Update `versions.txt`
 - Operator, target allocator and opamp-bridge should be `$VERSION`.
 - OpenTelemetry Collector should be the latest collector version. The `major.minor` should typically match, with the patch portion being possibly different.
 - The `autoinstrumentation-*` versions should match the latest supported versions in `autoinstrumentation/`.
- ⚠ Warning

DO NOT BUMP JAVA PAST `1.x.x` AND DO NOT BUMP .NET PAST `1.2.0`. Upgrades past these versions will introduce breaking HTTP semantic convention changes.

- iii. Check if the compatible OpenShift versions are updated in the `Makefile`.
- iv. Update the bundle by running `make bundle reset VERSION=$VERSION`.
- v. Change the compatibility matrix in the [compatibility doc](#). Remove the oldest entry.
 - Use the OpenTelemetry Operator version to be released

SemConv Migrations

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[chore][RFC] Semantic conventions migrations in the Collector #14273

Code ▾

Merged mx-psi merged 14 commits into open-telemetry:main from mx-psi:mx-psi/rfc-feature-gates on Jan 26

Conversation 44 Commits 14 Checks 83 Files changed 2

+239



mx-psi commented on Dec 9, 2025 • edited ▾

Member ...

Description

RFC about dealing with semantic conventions migrations.

This would be applicable to semantic conventions migrations related to RPC, system metrics, Kubernetes metrics and attributes...

RFC Checklist

- ☒ Announced on the Jan 14, 2026 Collector SIG meeting
- ☒ Mentioned in #otel-collector-dev <https://cloud-native.slack.com/archives/C07CCCMRXBK/p1768400424432639>



[chore][RFC] Semantic conventions migrations in the Collector

Verified



codecov bot commented on Dec 9, 2025 • edited ▾

<https://github.com/open-telemetry/opentelemetry-collector/pull/14273>
<https://github.com/open-telemetry/opentelemetry-collector-contrib/issues/44589>

Reviewers

Imolkova

dmathieu

jade-guiton-dd

codeboten

bogdandrutu

dmitryax

k8sattributes processor semantic conventions transition #44589

New issue

Closed Parent: Promote k8sattributes Processor to Stable Listed in #44483



ChrsMark opened on Nov 27, 2025 • edited by ChrsMark

Edits ▾ Member ...

Component(s)

processor/k8sattributes

Describe the issue you're reporting

Part of [\[k8s\] Transition to stable semconv #43869](#)

This is a placeholder issue to track the transition of this component to stable Semantic Conventions.

Probably this will happen after [#35325](#), following a similar approach.

Cross-posting from [open-telemetry/semantic-conventions#3120](#), the list of expected breaking changes:

- container.image.tag is a future breaking since it will need to become container.image.tags to follow [semconv](#)
- The k8s.*.labels/annotations.* breaking change is handled by [k8sattr.labelsAnnotationsSingular.allow](#) feature gate which's maturity is tracked at [\[k8sattributes\] Mature k8sattr.labelsAnnotationsSingular.allow feature gate #44693](#). This was also tracked by [Ensure that k8s labels and annotations follow SemConv #32440](#).
- k8s.pod.hostname, k8s.pod.ip, k8s.pod.start_time are not defined as Semantic Conventions yet.

Assignees

ChrsMark

Labels

component-stability-phase-1 never stale
priority:p1 processor/k8sattributes

Type

No type

Projects

Component Stability

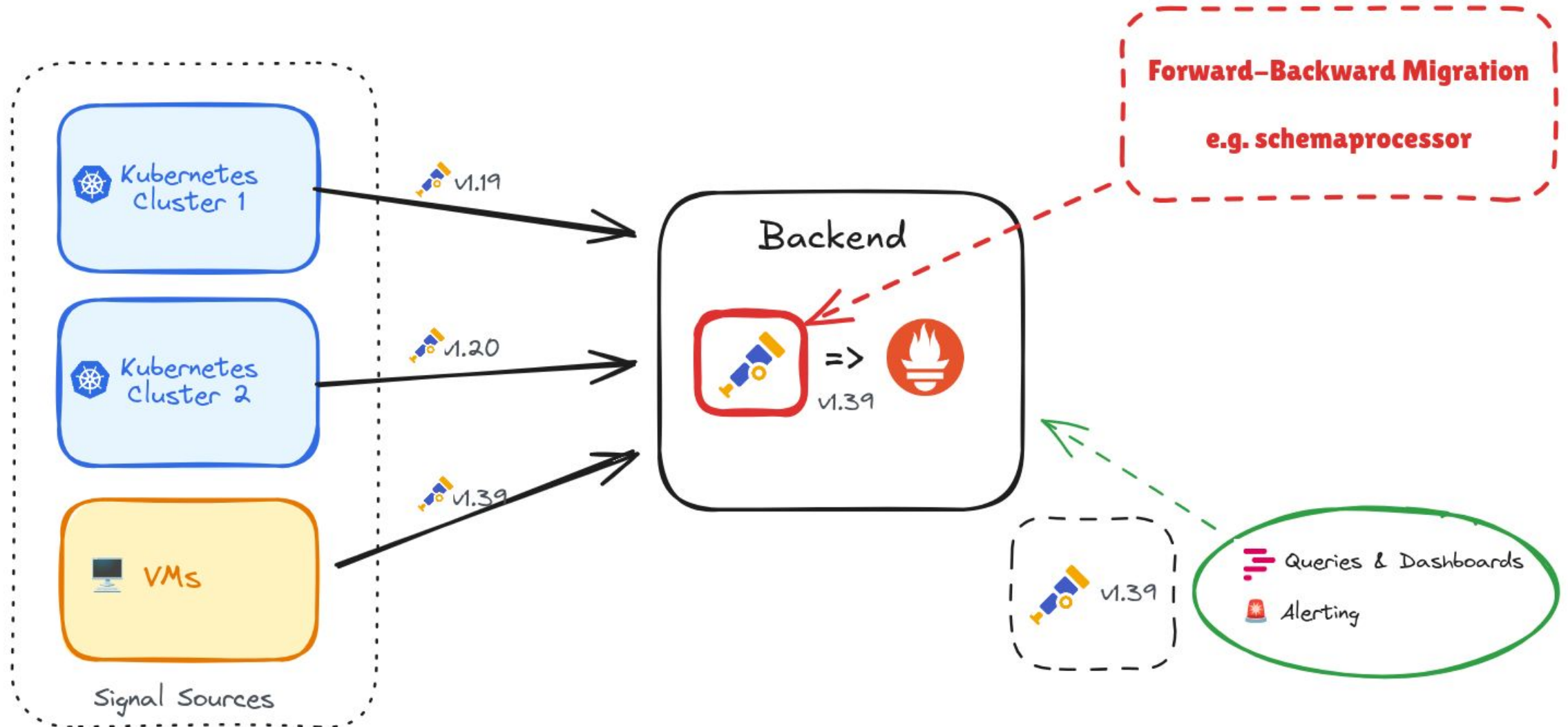
Status Done

Milestone

No milestone

Relationships

Breaking Changes - Ingestion



Forward compatibility

- Newer versions must be supported by the backend first
- Dashboards must be maintained and upgraded on a regular basis

Backward compatibility

- It is not feasible to expect constant migration back.
- If possible, capture what matters to you in a customizable format

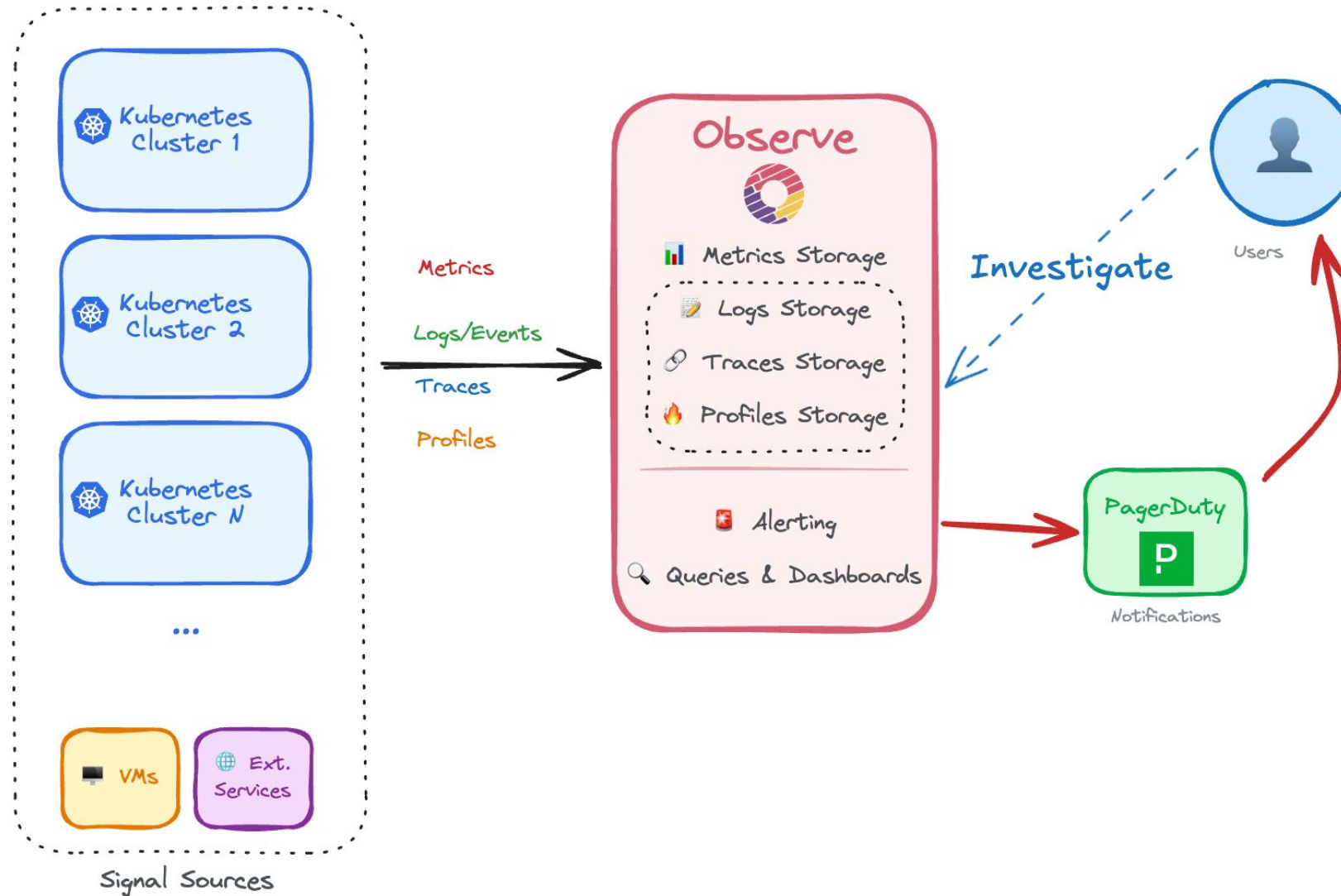
Where to Handle Schema Evolution?

Layer	Strengths	Trade-offs	Best fit
<i>Instrumentation-time</i>	Clean data at the source	Hard to coordinate across languages, SDKs, and teams	Homogeneous environments with strong ownership
<i>Ingest-time</i>	Centralized normalization; stable queries	Higher ingest complexity and cost	Platform teams operating shared pipelines
<i>Query-time</i>	Fastest to adopt; no producer changes required	Dashboards, alerts, and queries become more complex	Transitional periods and mixed-version environments

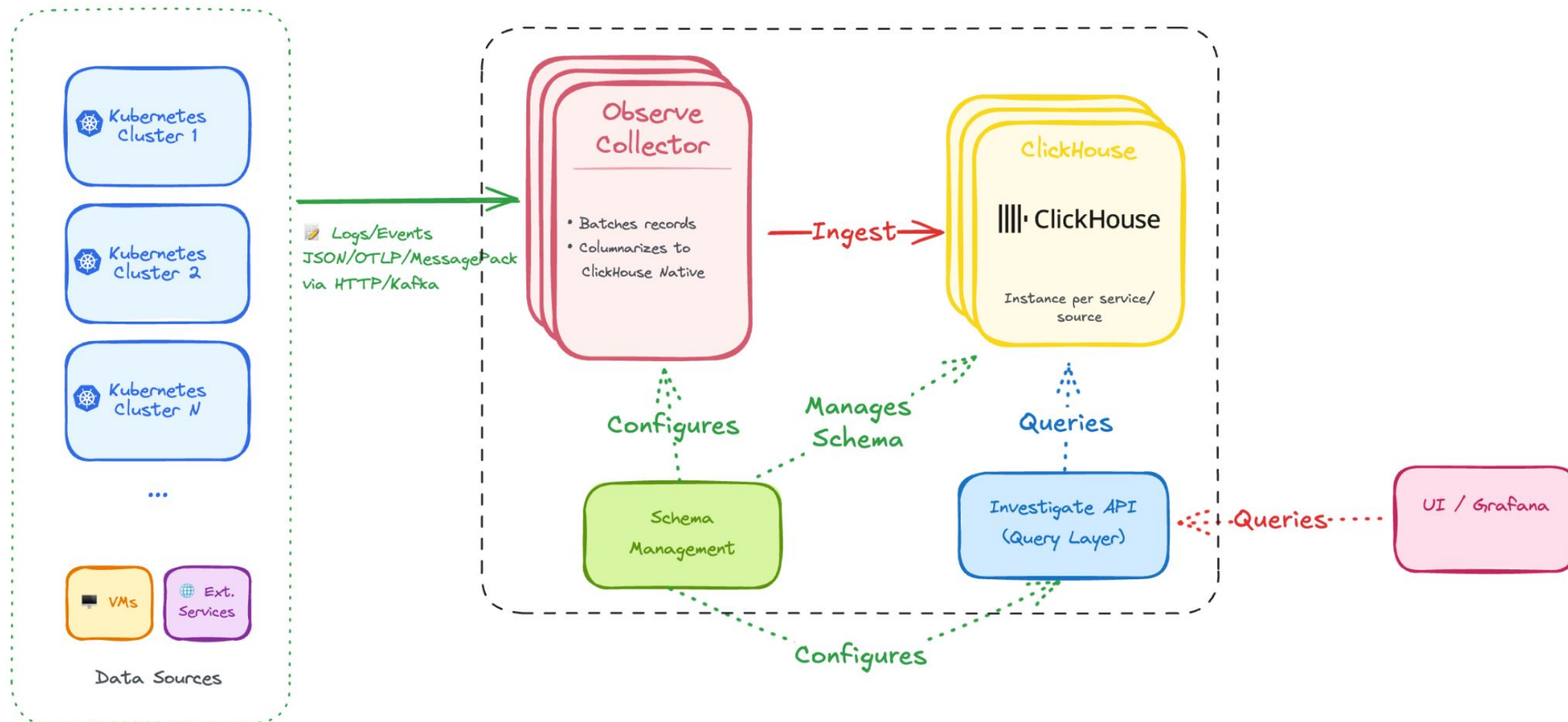
Shopify's Approach

Case Study: Evolving an Event Schema

Shopify Observe



Logs / Events Data Path



Database Schema for Physical Storage

```
CREATE TABLE IF NOT EXISTS logging.logs_v0
(
  `timestamp` DateTime64(9),

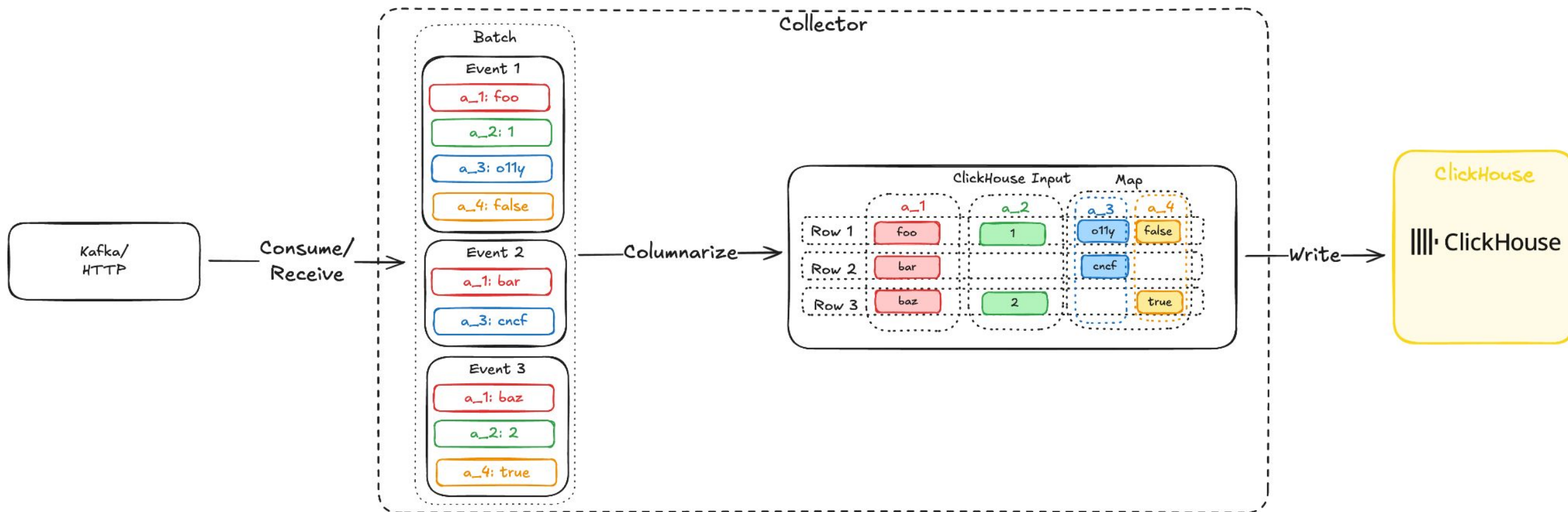
  -- promoted columns
  `request_id` String,
  `trace_id`   String,
  `component`  String,
  `message`    String,

  -- maps
  `string_attributes` Map(String, String),
  `int_attributes`    Map(String, Int64),
  `float_attributes`  Map(String, Float64),
  `bool_attributes`   Map(String, Bool)
)
ENGINE = MergeTree
PARTITION BY toStartOfHour(timestamp)
ORDER BY (component, toUnixTimestamp(timestamp));
```

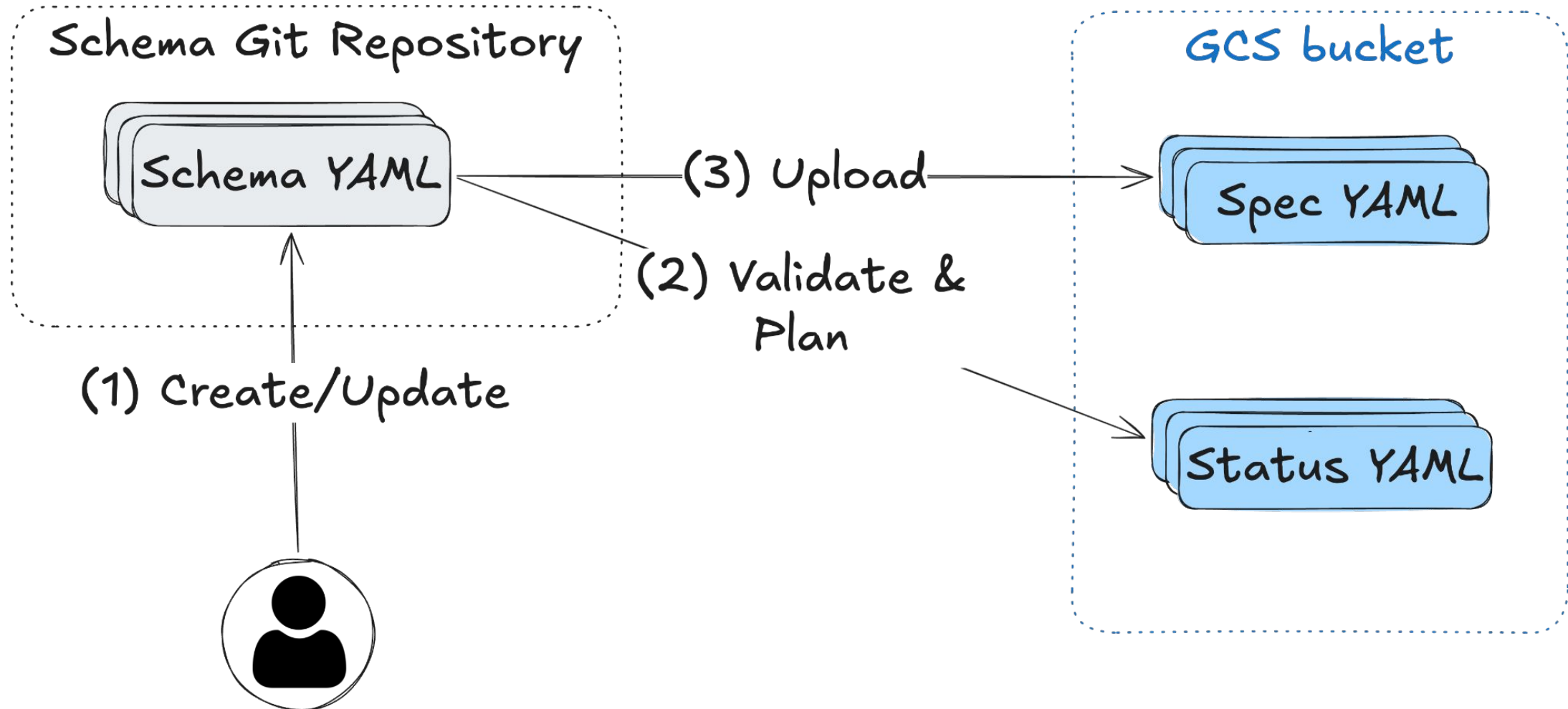
- **Column name**
- **Path(s) in the log/event payload**
- **Column type**

```
name: logs_v0
tenant: tenant_1
fields:
  - name: request_id
    path: context.request_id
    kind:
      type: string
  ...
```

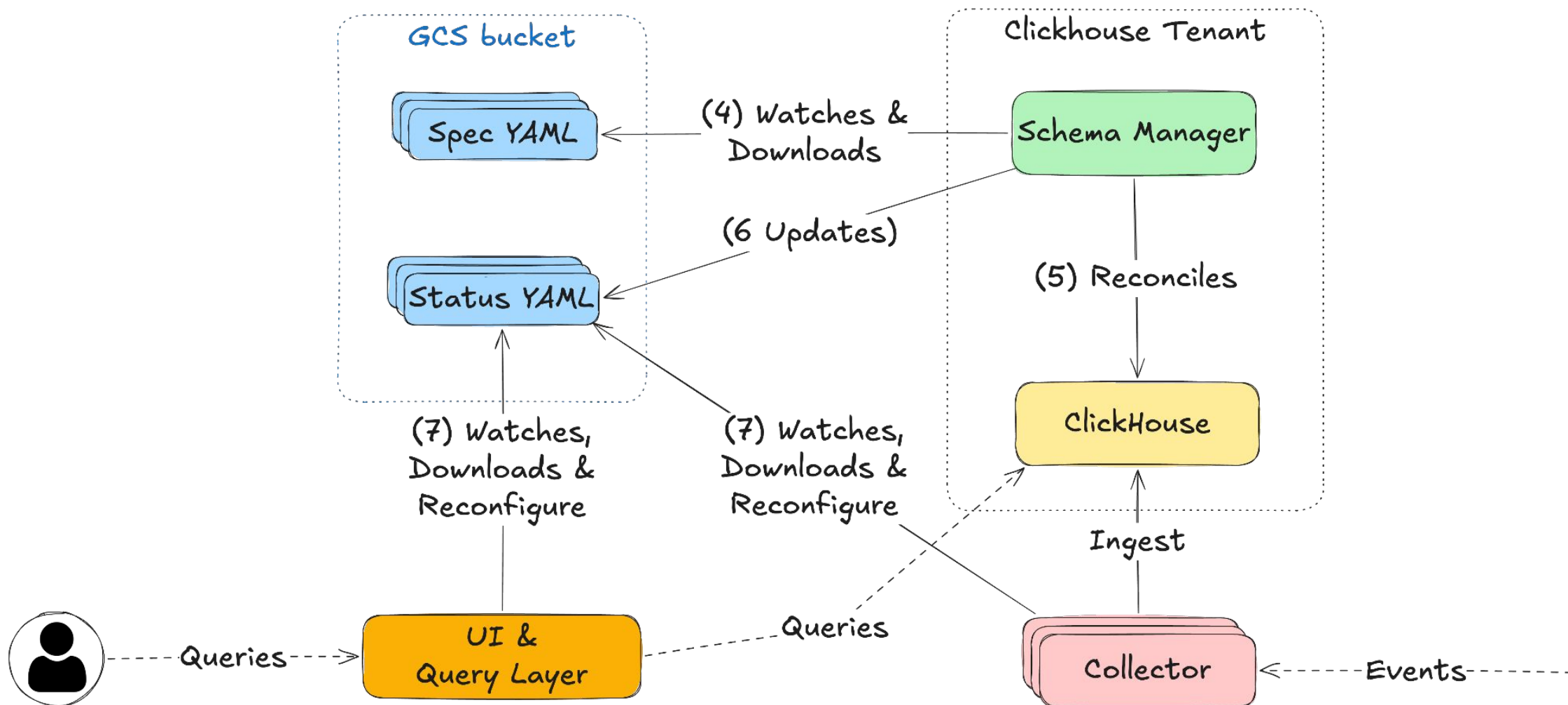
Ingestion



Schema Authoring and Distribution



Applying Schemas at Runtime



Example: Forward-Comp. Field Mapping

1. {"key": "http.status_code", "value": {"stringValue": "200"}}
2. {"key": "http.response.status_code", "value": {"stringValue": "200"}}
3. {"key": "http.response.status_code", "value": {"intValue": 200 }}

1. & 2.

```
name: logs_v0
tenant: tenant_1
fields:
  - name: http_response_status_code
    path: http.status_code
    kind:
      type: int
    ...
```



```
SELECT
  timestamp,
  http_response_status_code,
  string_attributes
FROM logging.otlp_v0
ORDER BY timestamp ASC
```

Query id: 04c54629-01c9-49d7-ac52-ae72f6870ee3

	timestamp	http_response_status_code	string_attributes
1.	2026-01-21 19:10:55.000000000	200	{}
2.	2026-01-21 19:11:07.000000000	NULL	{'http.response.status_code': '200'}
3.	2026-01-21 19:13:12.000000000	200	{}

3.

```
name: logs_v0
tenant: tenant_1
fields:
  - name: http_response_status_code
    paths:
      - http.status_code
      - http.response.status_code
    kind:
      type: int
    ...
```


- Define a canonical logical schema
- Version schema changes explicitly
- Preserve compatibility during migrations
- Choose one primary normalization layer
- Validate changes before rollout
- Keep dashboards and alerts aligned
- Measure progress and retire old mappings safely

The goal is not to avoid change, but to control it.

- Standardization is important
 - Enhances collaboration
- Standardization is hard
 - We have to deal with ongoing changes whether we like it or not
- Normalization needs to take place somewhere
 - At the instrumentation level
 - Ingestion time (Forward / Backward)
 - Query time
- One upcoming helper might be OpenTelemetry Weaver

OpenTelemetry Weaver



Tomorrow 15:15 CET,
by Arthur Silva Sens and Nicolas Takashi

<https://kccnceu2026.sched.com/event/2CVzR>

Thanks! Questions?

Feedback



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